

Talha Laique

Machine Learning Engineer | Computer Vision

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Ås, Norway | Eligible to work in Norway (Residence Permit – no sponsorship required)

Portfolio: <https://laiquet.github.io/>

PROFILE

Machine Learning / Computer Vision Engineer with 5+ years of experience developing and deploying computer vision, classification, and document AI systems across production and applied research environments. Experienced across the full ML lifecycle, including dataset creation and annotation, model development and evaluation (CNNs, YOLO, SAM), experiment tracking with MLflow, and deployment through Dockerized Flask/REST APIs. Proven ability to own end-to-end ML pipelines, from raw image and document data to production-ready inference services.

TECHNICAL SKILLS

Programming / Data: Python, SQL, NumPy, pandas, Matplotlib

Machine Learning / Deep Learning: PyTorch, TensorFlow, scikit-learn, CNNs, Ensemble Learning, Model Training & Evaluation

Computer Vision: YOLO (v8/v11/v26), SAM, Image Classification, Object Detection, Instance Segmentation, Image Annotation

Document AI: OCR, Document Understanding, Information Extraction, Document Classification

MLOps / Deployment: Docker, MLflow, FastAPI, Flask, Gunicorn, REST APIs, Git, Linux, GCP, Azure

LLM / Generative AI: Hugging Face, LangChain, ChromaDB, Ollama, Gemini API, Retrieval-Augmented Generation (RAG)

EXPERIENCE

NMBU | COMPUTER VISION & MACHINE LEARNING RESEARCHER (PHD CANDIDATE)

Ås, Norway | Jan 2024 –

Present

- Developed and validated end-to-end computer vision pipelines for Atlantic salmon phenotyping using YOLOv8/YOLOv11, SAM 2.1, projective registration, photometric normalization, and feature extraction; achieved 0.995 mAP@50 for operculum segmentation and 0.895 Dice for spot masks on held-out data.
- Built and quality-controlled instance-segmentation datasets with 275 annotated head images and ~9,000 polygon-labeled spots; expanded training data to 1,065 images (~35,000 spot instances) and benchmarked detection and segmentation models across train, validation, and test sets.
- Engineered longitudinal image-analysis workflows to quantify 2,615 melanin spots and track 676 matched pre/post-confinement spot pairs; introduced EXIF-based photometric normalization that corrected up to 2.7 stops of exposure variation, enabling robust pigmentation measurements across imaging sessions.

GIK INSTITUTE | LECTURER – ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Topi, Swabi, Pakistan | Apr 2022 –

Nov 2023

- Taught Machine Learning, Deep Learning, and Computer Vision courses with applied Python labs covering model training, evaluation, and data preprocessing.
- Designed AI lab material and workshops focused on practical use cases in classification, predictive modeling, and deep learning.

INIT ML | MACHINE LEARNING ENGINEER INTERN

Paris, France | May 2021 – Oct 2021

- Developed an end-to-end product taxonomy model to classify fashion items by category, sub-category, and multiple attribute levels using image and metadata inputs; deployed via a web application and API demo.
- Built web-scraping and data preprocessing pipelines to create a custom e-commerce dataset containing several thousand labeled product images for model training.
- Quantized and integrated the trained model into an Android application, enabling efficient on-device inference for product image classification.

BASIKON | MACHINE LEARNING ENGINEER INTERN – DOCUMENT AI

Paris, France | Apr 2020 – Aug 2020

- Developed a document-understanding pipeline to extract structured information from scanned financial documents across diverse layouts and varying scan quality.
- Built an internally annotated dataset of financial document images and prototyped CNN-based models for automated field extraction.

- Developed ensemble and embedding-based models to classify user career trajectories into 17 broad and 57 granular job categories using work-history and education data.
- Engineered features from professional experience, education, and semantic embeddings to improve classification performance across both levels of the job taxonomy.

PROJECTS

YOLOV8 SPOT DETECTOR - MLOPS PIPELINE

Python, Flask, YOLOv8, MLflow, Docker, Unicorn, REST API

- Deployed a Dockerized Flask API serving a custom YOLOv8 fish-spot detection model achieving 86.5% mAP@50, with MLflow for experiment tracking and model versioning.
- Developed an end-to-end inference application with authentication, rate limiting, and a drag-and-drop web interface for real-time object detection and bounding-box visualization.

SALMON YOLO PIPELINE - UNDERWATER FISH DETECTION

Python, YOLO26x, Ultralytics, OpenCV, Real-ESRGAN, CUDA, CLI

- Developed a 4-stage CLI pipeline (preprocess → train → infer → crop) integrating underwater image enhancement, YOLO-based detection, and ROI extraction across a 207-image, 1,750-instance dataset.
- Designed an underwater image-enhancement pipeline using Gray-World white balance, CLAHE, and Real-ESRGAN, improving UIQM by 80% and UCIQE by 176%.
- Fine-tuned YOLO26 for underwater object detection, achieving 0.756 mAP@50 and a 37% improvement over the raw-image baseline.

AI-ASSISTED COMPUTER VISION ANNOTATION TOOL

Python, PySide6, PyTorch, OpenCV, SAM, YOLO, COCO

- Developed a desktop annotation tool supporting polygons, bounding boxes, and oriented bounding boxes, with undo/redo, autosave, crash recovery, and SAM-/YOLO-assisted labeling.
- Implemented multi-format import/export for YOLO, COCO, JSON, and SAM RLE, enabling seamless interoperability across annotation, training, and inference pipelines.

AI DOCUMENT OCR & TABLE EXTRACTION

JavaScript, Node.js, Express, Gemini 2.5 Flash, Multimodal AI

- Built a full-stack app using Gemini 2.5 Flash to extract structured tables from multi-page financial-document PDFs, with automatic document classification.
- Built an editable results interface with inline cell correction and multi-format export for production-ready data.

SWARM PAPER ANALYZER - MULTI-AGENT RESEARCH INTELLIGENCE

Python, LangGraph, LangChain, ChromaDB, Gemini API, RAG

- Architected a 6-agent LangGraph swarm (Supervisor + 5 specialists) with a fault-tolerant arXiv ingestion pipeline and ChromaDB RAG store for semantic retrieval.

EDUCATION

PhD - Computer Vision for Atlantic Salmon Welfare

Ås, Norway | Jan 2024 – Present

NMBU

Coursework: Computer Vision, Image Analysis, Animal Stress Biology

MScT - Artificial Intelligence and Advanced Visual Computing

Paris-Saclay, France | Sept 2021

UNIVERSITÉ PARIS-SACLAY – 3.67/4.0

Coursework: Computer Vision, Deep Learning, Machine Learning, NLP, Reinforcement Learning

BSc - Computer Engineering

Topi, Swabi, Pakistan | June 2015

GIK INSTITUTE – 3.22/4.0

Coursework: Data Mining, Databases, Data Structures & Algorithms, Digital Image Processing

PUBLICATIONS

- Ijaz, A. Z., Ali, R. H., Ali, N., **Laique, T.**, and Khan, T. A. *Solving Graph Coloring Problem via Graph Neural Network (GNN)*. ICET 2022, pp. 178–183.
- Lewis, R., Kostermans, T., Brovold, J. W., **Laique, T.**, and Ocepek, M. *Automated Body Condition Scoring in Dairy Cows Using 2D Imaging and Deep Learning*. AgriEngineering, 7(7), 241, 2025.
- **Laique, T.**, Gunnes, M., Folkedal, O., Nilsson, J., Green, E. A. L., Gundersen, H. N., Øverli, Ø., and Ullah, H. *Quantification of Opercular Pigmentation Changes in Farmed Atlantic Salmon: A Novel Application for Computer Vision in Fish Welfare Assessment*. Fishes, 11(5), 271, 2026.
- **Laique, T.**, Gunnes, M., Øverli, Ø., and Ullah, H. Computer Vision-Based Phenotyping of Melanin-Based Opercular Spot Architecture in Farmed Atlantic Salmon: Toward Image-Derived Welfare Monitoring. *Journal of Animal Science*. (Under review)
- **Laique, T.**, Gunnes, M., Øverli, Ø., and Ullah, H. Photometrically Controlled Quantification of Opercular Melanin Spot Fading in Confined Atlantic Salmon. *Computers and Electronics in Agriculture*. (Submitted)